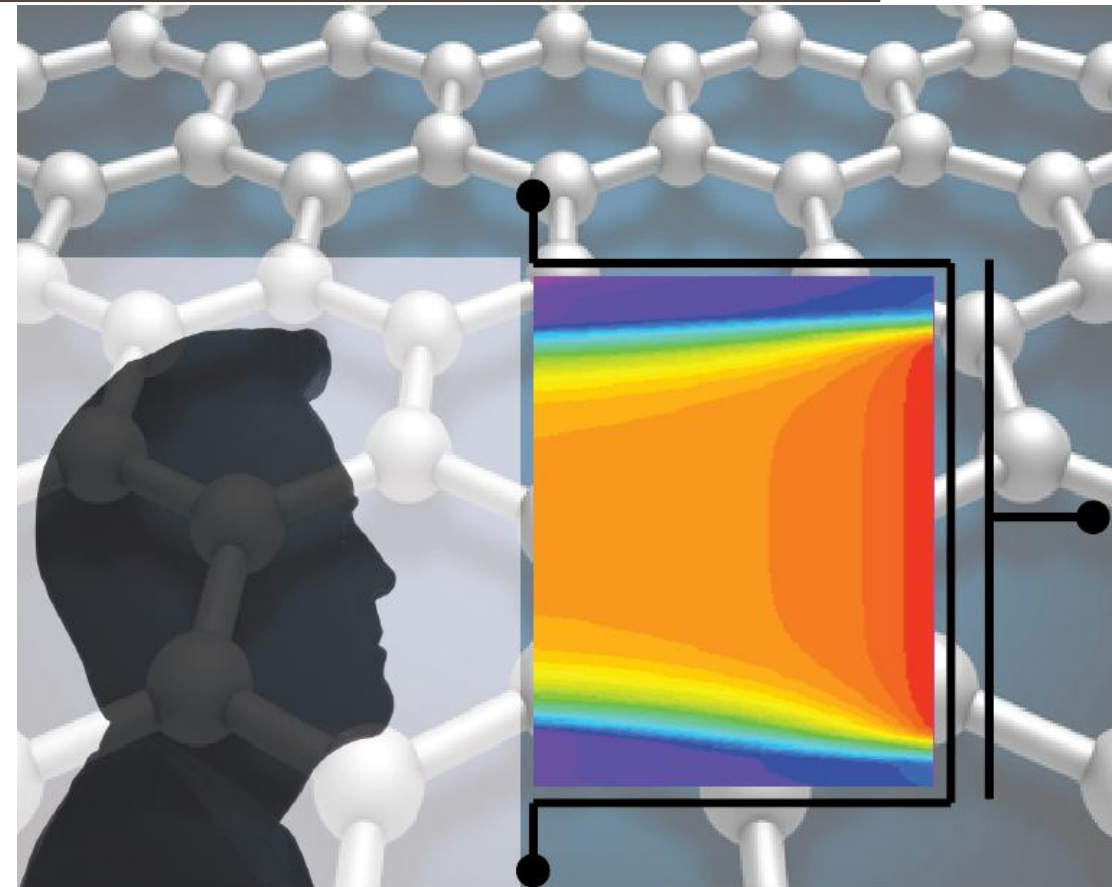


ADVANCED SOLID-STATE DEVICES

Week-9, Lecture-1

Sneh Saurabh
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MOSFET



Current-Voltage Characteristics

Current-Voltage Characteristics: Assumptions

- We have discussed electrical behavior of four-terminal MOSFET qualitatively
- Now, we will discuss them quantitatively

Assumptions:

1. Mobility is constant and independent of electric field (bias)
 - Valid only when electric field due to drain is less than 2-3 kV/cm
 - Not valid in modern devices, will relax this assumption later
2. Gradual channel approximation: gate to substrate field is much stronger than drain to source field
 - Valid when gate length is larger than the channel depth
 - Allows us to treat device using 1-D electric field rather than 2-D electric field

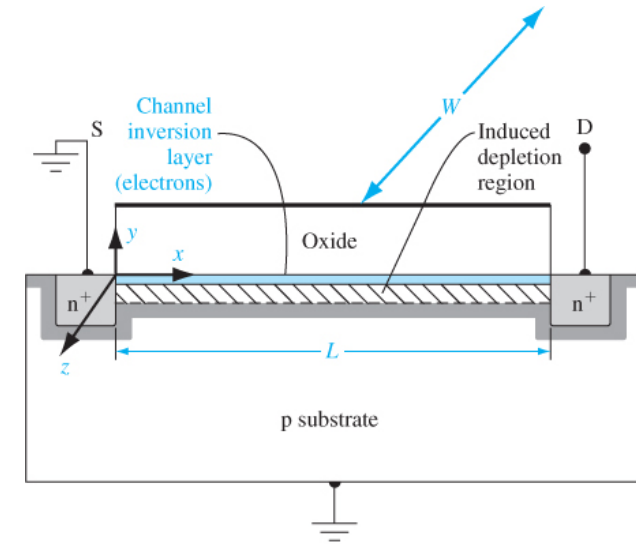


Figure 10.43 | Geometry of a MOSFET for I_D versus V_{DS} derivation.

Goal:

Find drain current in terms of drain voltage, gate voltage, and threshold voltage (assuming source is grounded) and MOSFET is in strong inversion

Current-Voltage Characteristics: Charge in the Inversion Region

- Potential balance: $V_{GS} = \phi_{ms} + \psi_s + \psi_{ox}$
- Charge balance: $Q'_G + Q'_{ss} + Q'_s = 0$

- We will use the above equations
- Semiconductor charge Q'_s consists of charge Q'_D in the depletion region below the channel and charge Q'_I in the inversion region:
- $Q'_s = Q'_I + Q'_D$
- First, we want to find Q'_I in terms of gate voltage, surface potential, and threshold voltage

From charge balance:

$$Q'_G = -(Q'_{ss} + Q'_I + Q'_D)$$

Voltage across oxide: $\psi_{ox} = \frac{Q'_G}{C'_{ox}} = \frac{-(Q'_{ss} + Q'_I + Q'_D)}{C'_{ox}}$

- From potential balance:

$$V_{GS} = \phi_{ms} + \psi_s - \frac{(Q'_{ss} + Q'_I + Q'_D)}{C'_{ox}}$$

$$V_{GS} = \left[\phi_{ms} - \frac{(Q'_{ss} + Q'_D)}{C'_{ox}} + 2\phi_{Fp} \right] - 2\phi_{Fp} + \psi_s - \frac{Q'_I}{C'_{ox}}$$

$$V_{GS} = V_T - 2\phi_{Fp} + \psi_s - \frac{Q'_I}{C'_{ox}}$$

So, $Q'_I = C'_{ox} [V_{GS} - V_T - (\psi_s - 2\phi_{Fp})]$

Current-Voltage Characteristics: Impact of Drain on the Inversion Region Charge

$$Q'_I = C'_{ox} [V_{GS} - V_T - (\psi_s - 2\phi_{Fp})]$$

- When drain is biased to a positive voltage, the surface potential becomes a function of x (position)

➤ $\psi_s(x)$ increases with x

- The reverse bias across the channel-substrate junction also increases with x

- Let $V(x)$ represent the channel potential for a drain voltage V_{DS} at position x

- We know:

➤ $V(x) = 0$ at $x = 0$ (at source)

➤ $V(x) = V_{DS}$ at $x = L$ (at drain)

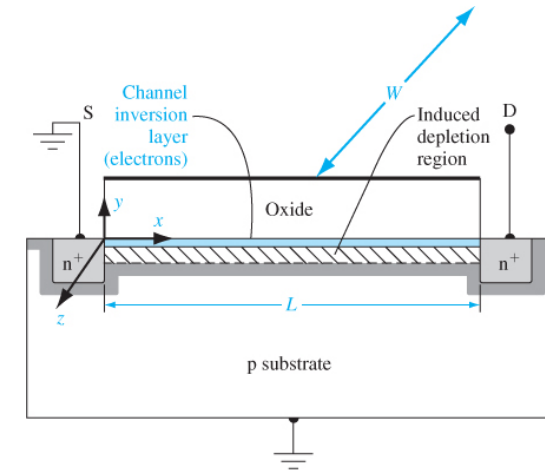


Figure 10.43 | Geometry of a MOSFET for I_D versus V_{DS} derivation.

- The surface potential in strong inversion can be approximated as:

$$\psi_s(x) = 2\phi_{Fp} + V(x)$$

We can write:

$$Q'_I = C'_{ox} [V_{GS} - V_T - V(x)]$$

Current-Voltage Characteristics: Drain Current

$$Q'_I = C'_{ox}[V_{GS} - V_T - V(x)]$$

We can write drain current due to drift as:

Drain current = Surface charge density $\times$ mobility $\times$ electric field $\times$ gate width

$$I_D = Q'_I \times \mu_n \times \frac{dV(x)}{dx} \times W$$

$$I_D dx = Q'_I \mu_n W dV(x)$$

$$\int_0^L I_D dx = \int_0^{V_{DS}} \mu_n W C'_{ox} [V_{GS} - V_T - V(x)] dV(x)$$

Current is constant throughout the device
(independent of x)

$$I_D L = \mu_n W C'_{ox} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$I_D = \frac{\mu_n W C'_{ox}}{L} V_{DS} \left[V_{GS} - V_T - \frac{V_{DS}}{2} \right]$$

Current-Voltage Characteristics: Operation Mode

$$Q'_I = C'_{ox}[V_{GS} - V_T - V(x)]$$

$$I_D = \frac{\mu_n W C'_{ox}}{L} V_{DS} \left[V_{GS} - V_T - \frac{V_{DS}}{2} \right]$$

When does Q'_I becomes zero at the drain end ($x = L$)?

At $x = L$, $V(x) = V_{DS}$

So, $C'_{ox}[V_{GS} - V_T - V_{DS}] = 0$

$$\Rightarrow V_{DS} = V_{GS} - V_T$$

- The inversion charge reduces to zero at the drain end when $V_{DS} = V_{GS} - V_T$ (This is known as the *pinch off condition*)

Linear Region (Ohmic): $V_{DS} < V_{GS} - V_T$

$I_D \approx \frac{\mu_n W C'_{ox}}{L} V_{DS} [V_{GS} - V_T]$ [Ignoring quadratic term for small V_{DS}]

Saturation Region: $V_{DS} \geq V_{GS} - V_T$

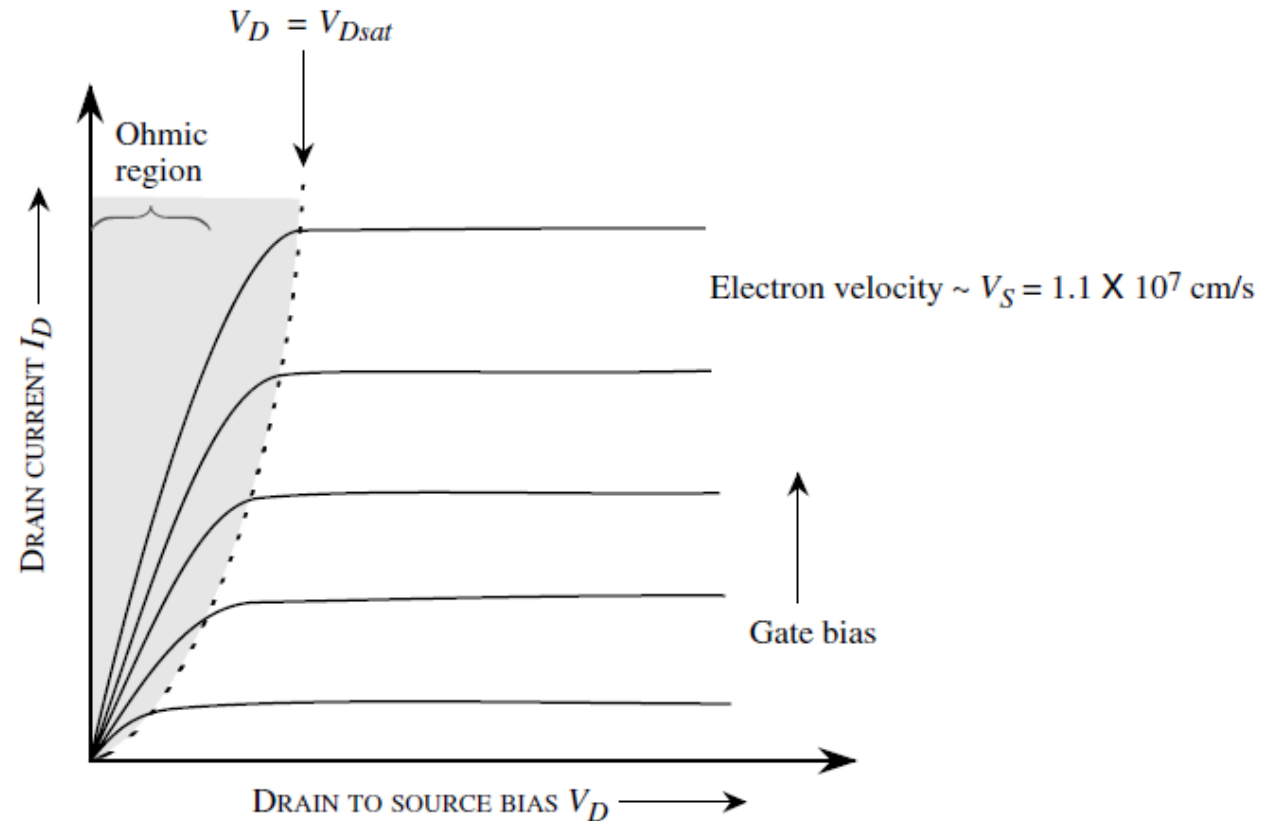
$$I_D = \frac{\mu_n W C'_{ox}}{2L} [V_{GS} - V_T]^2$$

- Current does not increase with drain voltage in saturation region
- Saturation current is quadratic in gate voltage and independent of drain voltage

Current-Voltage Characteristics: Output Characteristics

Output characteristics:
behaviour of drain current
for different V_{DS} for given
 V_{GS}

The spacing in the curve
increases quadratically with
increasing gate bias



Current-Voltage Characteristics: Transfer Characteristics

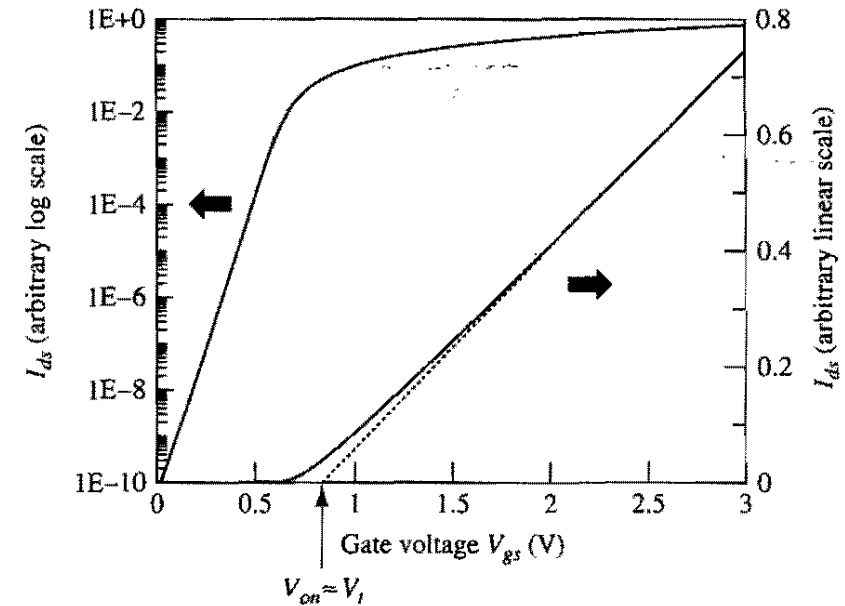
Transfer characteristics:
behaviour of drain current for
different V_{GS} for given V_{DS}

$$I_D = \frac{\mu_n W C'_{ox}}{L} V_{DS} \left[V_{GS} - V_T - \frac{V_{DS}}{2} \right]$$

Drain voltage is assumed to be small: $I_D \approx$

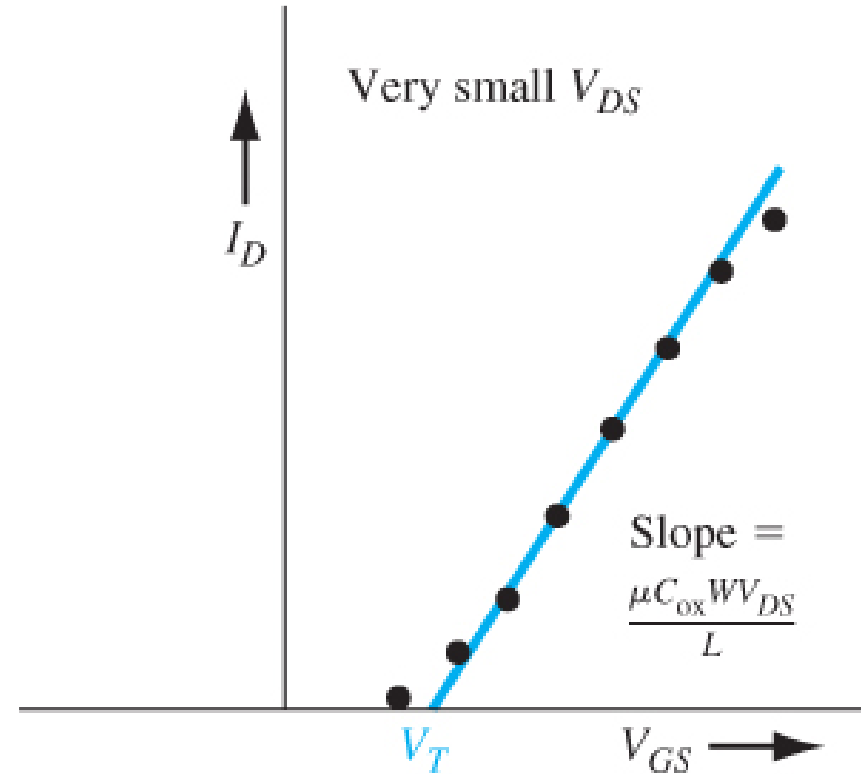
$$\frac{\mu_n W C'_{ox}}{L} V_{DS} [V_{GS} - V_T]$$

- $I_D = 0$ when $V_{GS} < V_T$ [Ideally, but there exists subthreshold current, which we will discuss later]
- $I_D \approx \frac{\mu_n W C'_{ox}}{L} V_{DS} [V_{GS} - V_T]$ when $V_{GS} > V_T$



Current-Voltage Characteristics: Extraction of Threshold Voltage

- Measure and plot transfer characteristics for a MOSFET
- Threshold voltage and mobility can be extracted using it.



MOSFET: Problem

Problem: Consider an ideal MOSFET with the following parameters:

$$L = 1.25 \mu m$$

$$\mu_n = 650 \text{ cm}^2/\text{Vs}$$

$$C_{ox}' = 6.9 \times 10^{-8} \text{ F/cm}^2$$

$$V_T = 0.65 \text{ V}$$

Design the channel width W such that the saturation current is 4 mA at $V_{GS} = 5 \text{ V}$

Solution:

$$I_D = \frac{\mu_n W C_{ox}'}{2L} [V_{GS} - V_T]^2$$

Answer:

$$W = 11.8 \mu m$$

Performance Measures: Drain conductance

- Drain conductance definition $g_d = \left. \frac{\partial I_D}{\partial V_{DS}} \right|_{V_{GS}=\text{constant}}$

- When drain bias is low: $g_d = \frac{\partial \frac{\mu_n W C'_{ox}}{L} V_{DS} [V_{GS} - V_T]}{\partial V_{DS}}$
$$= \frac{\mu_n W C'_{ox}}{L} [V_{GS} - V_T]$$

- In saturation, $g_d = 0$

- Low g_d is desirable for MOSFET when used as constant current source
- In real devices $g_d = 0$ may not be achievable (as we will see later)

Performance Measures: Transconductance

- Transconductance definition:

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS}=\text{constant}}$$

- When drain bias is low:

$$\begin{aligned} g_m &= \frac{\partial \frac{\mu_n W C'_{ox}}{L} V_{DS} [V_{GS} - V_T]}{\partial V_{GS}} \\ &= \frac{\mu_n W C'_{ox}}{L} V_{DS} \end{aligned}$$

- In saturation, $g_m = \frac{\mu_n W C'_{ox}}{L} (V_{GS} - V_T)$

- Transconductance closely related to speed of MOSFET (desired to be high)
- To get high g_m :
 - High mobility
 - Large W/L
 - Small thickness of gate oxide
 - Larger gate overdrive ($V_{GS} - V_T$)

MOSFET



Frequency Limitations

MOSFET: Small-Signal Model

- MOSFETs are used in analog circuits as amplifiers
 - Small signal model is required to mathematically analyze the circuit
- Small signal model assumes that the DC bias is already existing
 - Typically, MOSFET is biased in saturation for amplifier operation
 - Small signal model contains circuit elements (resistors and capacitors) that will be relevant in observing the response when the MOSFET is subjected to a small AC signal superimposed over the DC signal
- Frequency response is how the output AC signal will behave with respect to input AC signal at different frequencies
 - We will derive this frequency response with the help of small signal model

Small-Signal Model: Circuit Components

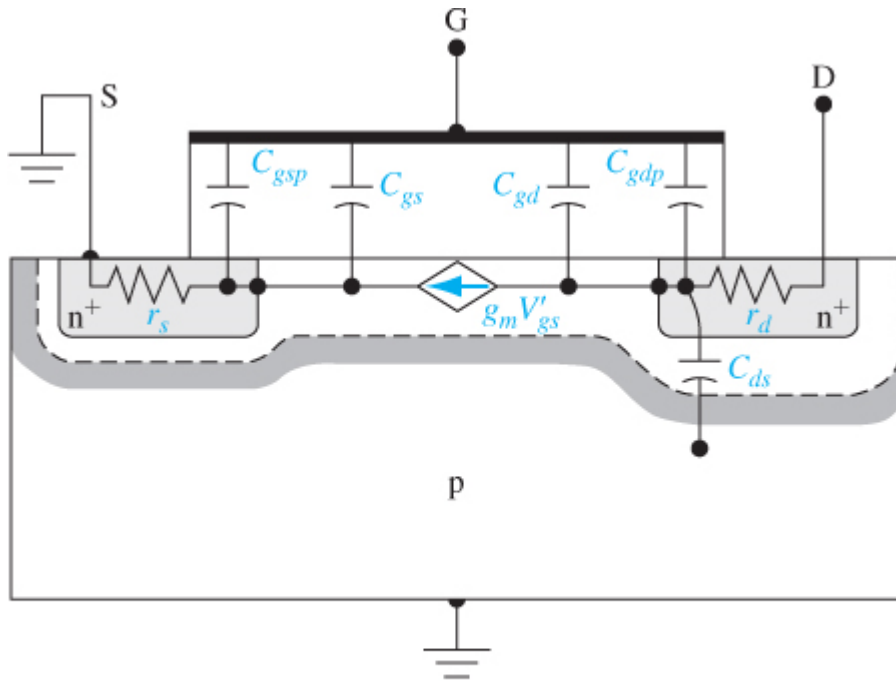


Figure 10.52 | Inherent resistances and capacitances in the n-channel MOSFET structure.

Capacitances associated with Gate Terminal:

- C_{gs}/C_{gd} : interaction between gate and channel charge near source/drain
- C_{gsp}/C_{gdp} : overlap capacitance between gate and source/drain
- C_{ds} : drain-to-substrate PN junction capacitance
- $g_m V'_gs$: voltage-controlled current source
- r_s/r_d : source/drain series resistance

Small-Signal Model: Equivalent Circuit

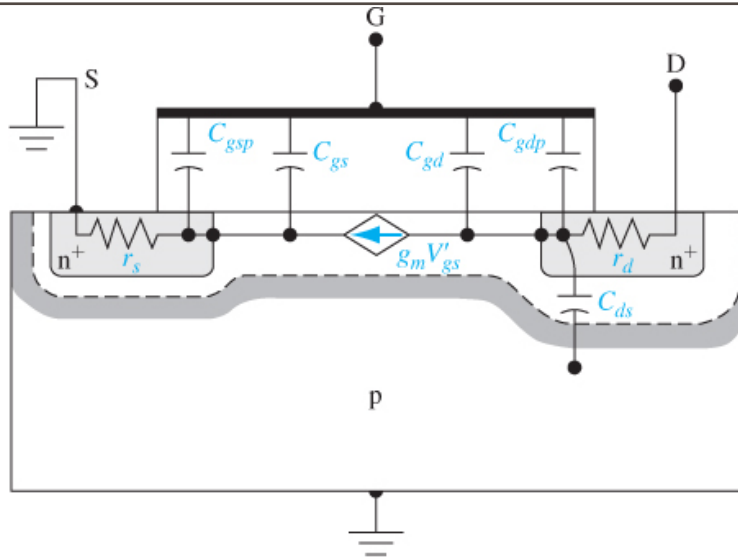


Figure 10.52 | Inherent resistances and capacitances in the n-channel MOSFET structure.

Translation to an equivalent circuit is straightforward

- C_{gsT} : total gate-to-source capacitance ($C_{gsT} = C_{gsp} + C_{gs}$)
- C_{gdT} : total gate-to-drain capacitance ($C_{gdT} = C_{gdp} + C_{gd}$)

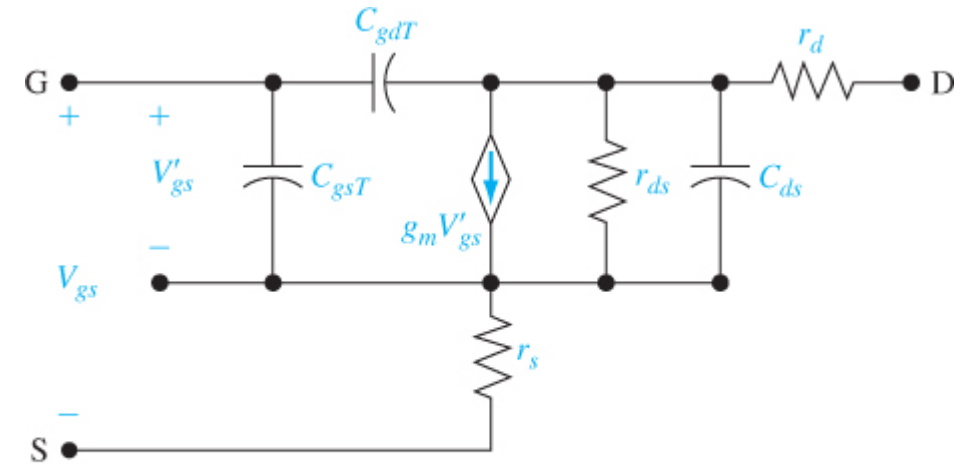


Figure 10.53 | Small-signal equivalent circuit of a common-source n-channel MOSFET.

V'_{gs} : internal gate-to-channel voltage that controls channel current

- $r_{ds} = \frac{1}{g_d}$ [g_d is the drain conductance that was computed earlier]
- r_{ds} is infinity in saturation (ideally)

Small-Signal Model: Equivalent Circuit (Simplified)

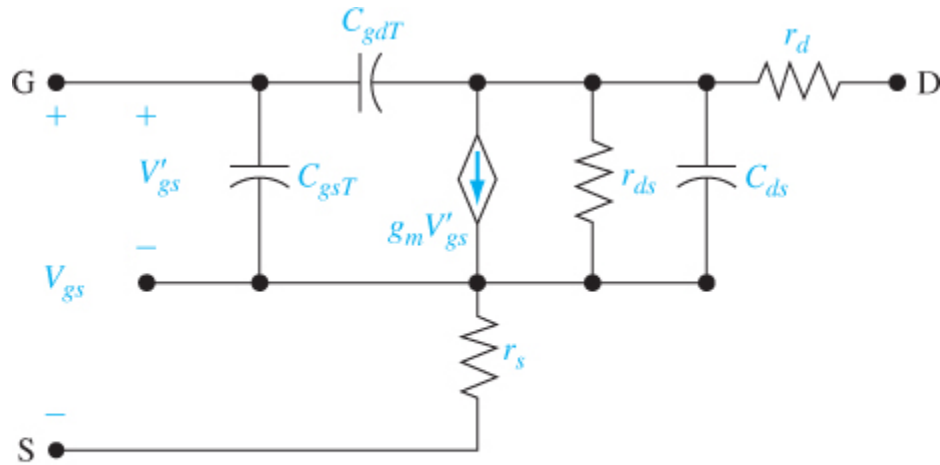


Figure 10.53 | Small-signal equivalent circuit of a common-source n-channel MOSFET.

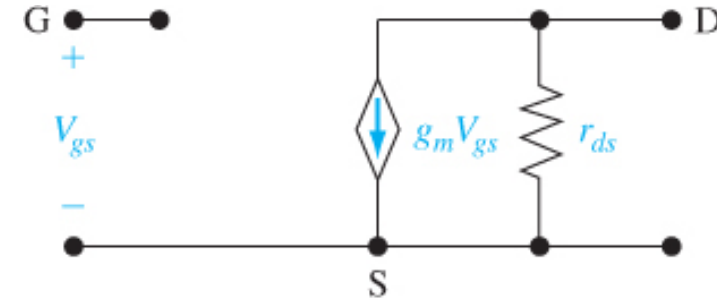


Figure 10.54 | Simplified, low-frequency small-signal equivalent circuit of a common-source n-channel MOSFET.

Equivalent circuit can be simplified as follows:

- $r_s = r_d = 0$
- Input gate impedance very high (C_{gsT} and C_{gdT} open circuited) and C_{ds} assumed to be insignificant

Small-Signal Model: Impact of Source Resistance

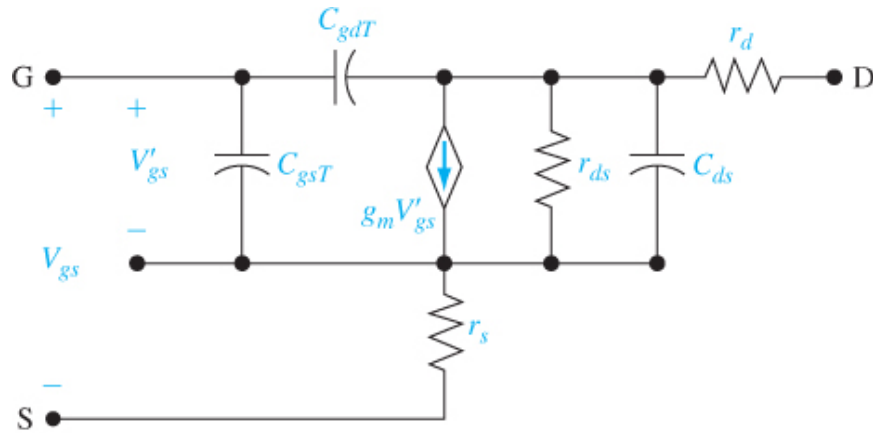


Figure 10.53 | Small-signal equivalent circuit of a common-source n-channel MOSFET.

- Source resistance can be small, still it can show greater impact
- It can be understood from a simplified circuit in which we keep r_s and remove all other components, except the current source.

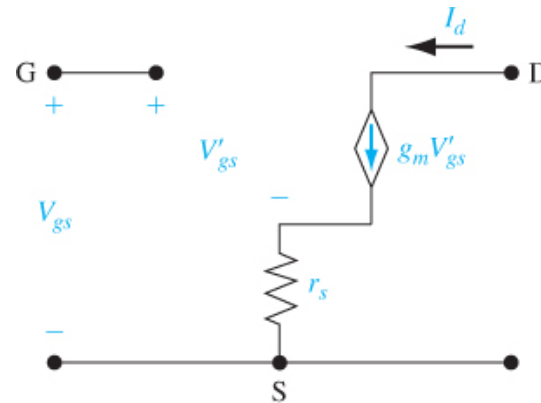


Figure 10.55 | Simplified, low-frequency small-signal equivalent circuit of common-source n-channel MOSFET including source resistance r_s .

Problem: Express drain current I_d in terms of V_{gs}

$$\begin{aligned} I_d &= g_m V'_{gs} \\ V_{gs} &= V'_{gs} + I_d r_s \\ &= \frac{I_d}{g_m} + I_d r_s \\ &= \frac{I_d (1 + g_m r_s)}{g_m} \end{aligned}$$

$$I_d = \frac{g_m}{(1 + g_m r_s)} V_{gs}$$

- Source resistance reduces the overall transistor gain (since g_m is large, the product $g_m r_s$ can be large)

- Small r_s can also have significant impact

References

- Tsividis, Yannis. *Operation and Modeling of the MOS Transistor*. McGraw-Hill, Inc., 1987.
- D. A. Neaman, "*Semiconductor Physics and Devices*"